Murtaza Ahmed

Undergrad Engineer | Distributed Systems, ML Systems & Infra (Infrastructure focus)

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I'm a systems engineer and undergraduate researcher focused on building the future of AI infrastructure. I design resilient cluster orchestration policy engines, optimize edge computer vision pipelines, and build custom hardware platforms. Welcome to my digital workspace—feel free to check out my latest research, open-source projects, or download my typeset resumes!

EDUCATION

Bachelor of Engineering in Mechanical Engineering — **May 2028**
Muffakham Jah College of Engineering & Technology (Osmania University) — *Hyderabad, TG*

- Relevant Coursework: Scientific Computing, Embedded Systems Programming, System Modeling & Simulation

EXPERIENCE

Undergraduate Research Assistant (Edge AI Systems) — **Dec 2025 -- Present**
Department of Mechanical Engineering (MJCET) — *Hyderabad, TG*

- Selected for an institution-funded R&D program under faculty supervision.
- Built a complete edge AI system on NVIDIA Jetson Nano 4GB Developer Kit, integrating an IMX719 camera and soil sensors for real-time field operation.
- Developed the on-device inference pipeline and executed real-time inference under compute and memory constraints.
- Achieved 40–70 ms latency (14–20 FPS) using TensorRT (FP16) optimization.
- Processed camera and sensor data continuously at 5–10 Hz with low-latency performance.
- Designed a stable execution loop enabling reliable operation over 4–6 hour field deployments.

Undergraduate Researcher (R&D Program) — **March 2026 -- Present**
Department of Mechanical Engineering (MJCET) — *Hyderabad, TG*

- Design and implementation of an Adaptive Multi-Objective Policy Engine for Resilient Orchestration in Heterogeneous LLM Clusters.
- Researching adaptive workload scheduling and hardware-aware resource allocation strategies for heterogeneous LLM inference systems.
- Exploring fault-tolerant orchestration approaches for distributed AI workloads across mixed compute environments.

Builder & Systems Architect — **July 2025 -- Present**
Verity (Decentralized Verification Platform) — *Hyderabad, TG*

- Designing and implementing a system to verify digital media provenance using blockchain and decentralized storage.
- Integrating AI modules for manipulation detection and automated claim extraction.
- Developing media provenance infrastructure using distributed storage systems including IPFS and Arweave.
- Building concurrent ingestion pipelines and immutable audit logging mechanisms for ML-based verification workflows.

Backend Infrastructure Engineer — **Sept 2024 -- Present**
Freelance Consulting — *Hyderabad, TG*

- Migrated client applications to containerized Linux deployment environments with automated CI/CD workflows.
- Designed backend APIs and optimized PostgreSQL schemas for improved concurrency and deployment reliability.

PROJECTS

Smart Agri Four-Legged Bot (SAFL-B) | C • Python • TensorRT • NVIDIA Jetson Nano • Arduino Due • SolidWorks • FEA • UART/Serial — **May 2026**

- Awarded 1st Place at the city-wide MAKEFORHYDERABAD Makeathon (organized by Titan & SIT Hyderabad) for autonomous robotics integration.
- Designed a hybrid 3mm steel chassis and custom 3D-printed modular articulated planar legs featuring passive compliance and internal wire routing channels.

- Validated structural load paths using Finite Element Analysis (FEA) under static vertical loads, bumps, and probe insertions with a minimum safety factor of 2.04.
- Integrated a local CNN weed classification model on NVIDIA Jetson Nano, achieving real-time inference (9.2 FPS) using TensorRT FP16 acceleration.
- Designed low-level kinematic control systems using an Arduino Due to orchestrate H-Bridge DC motor actuators and high-current ESCs.

Deterministic Workspace Provisioning | Bash • Linux • Arch Linux • Podman • Docker • Git **June 2025**

- Built automated provisioning scripts for lightweight developer workstation deployment using Arch Linux and custom dotfiles.
- Configured layered build pipelines to build reproducible container environments, separating high-current ML layers from system packages.
- Automated dependency setups, window manager configurations (i3wm), and performance optimizations for Intel/NVIDIA drivers.

TOOLS & METHODS

Languages: C, Python, Bash, PostgreSQL, Git
Systems: Linux, Arch Linux, Docker, Kubernetes, Podman